

Double Data Encryption Standard (2DES)

Aim

To implement the **Double Data Encryption Standard (2DES)** algorithm for encrypting and decrypting plaintext using **two different secret keys**, thereby improving security over the standard DES algorithm.

Algorithm

Encryption Algorithm

Step 1: Start the program.

Step 2: Read the plaintext from the user.

Step 3: Read the first secret key (K1).

Step 4: Read the second secret key (K2).

Step 5: Encrypt the plaintext using the first key (K1) with DES.

$$C_1 = DES_{K1}(P)$$

Step 6: Encrypt the intermediate ciphertext again using the second key (K2).

$$C = DES_{K2}(C_1)$$

Step 7: Display the final ciphertext.

Step 8: For decryption, first decrypt the ciphertext using the second key (K2).

$$C_1 = DES_{K2}^{-1}(C)$$

Step 9: Decrypt the intermediate ciphertext using the first key (K1).

$$P = DES_{K1}^{-1}(C_1)$$

Step 10: Display the original plaintext.

Step 11: Stop the program.

C Program (Simplified 2DES Demonstration)

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char plaintext[100], key1[100], key2[100];
```

```
    char cipher1[100], cipher2[100], decrypted[100];
```

```

int i, len;

printf("Enter Plaintext: ");
scanf("%s", plaintext);
printf("Enter First Key: ");
scanf("%s", key1);
printf("Enter Second Key: ");
scanf("%s", key2);
len = strlen(plaintext);

// First Encryption using Key1
for(i = 0; i < len; i++)
{
    cipher1[i] = plaintext[i] ^ key1[i % strlen(key1)];
}

// Second Encryption using Key2
for(i = 0; i < len; i++)
{
    cipher2[i] = cipher1[i] ^ key2[i % strlen(key2)];
}

printf("\nEncrypted Cipher Text: ");
for(i = 0; i < len; i++)
{
    printf("%02X", (unsigned char)cipher2[i]);
}

// First Decryption using Key2
for(i = 0; i < len; i++)
{
    cipher1[i] = cipher2[i] ^ key2[i % strlen(key2)];
}

// Second Decryption using Key1
for(i = 0; i < len; i++)
{
    decrypted[i] = cipher1[i] ^ key1[i % strlen(key1)];
}

```

```

    }

    decrypted[len] = '\0';

    printf("\nDecrypted Plaintext: %s\n", decrypted);

    return 0;

}

```

OUTPUT :

The screenshot shows a C++ IDE with the following components:

- Source Code (DES.cpp):**

```

1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char plaintext[100], key1[100], key2[100];
7     char cipher1[100], cipher2[100], decrypted[100];
8     int i, len;
9
10    printf("Enter Plaintext: ");
11    scanf("%s", plaintext);
12
13    printf("Enter First Key: ");
14    scanf("%s", key1);
15
16    printf("Enter Second Key: ");
17    scanf("%s", key2);
18
19    len = strlen(plaintext);
20
21    // First Encryption using Key1
22    for(i = 0; i < len; i++)
23    {
24        cipher1[i] = plaintext[i] ^ key1[i % strlen(key1)];
25    }
26
27    // Second Encryption using Key2
28    for(i = 0; i < len; i++)
29    {
30        cipher2[i] = cipher1[i] ^ key2[i % strlen(key2)];
31    }
32
33    printf("\nEncrypted Cipher Text: ");
34    for(i = 0; i < len; i++)
35    {
36        printf("%02X", (unsigned char)cipher2[i]);
37    }
38
39    // Decryption (omitted in the provided code block)
40
41    return 0;
42 }

```
- Terminal Output:**

```

Enter Plaintext: HELLO
Enter First Key: KEY1
Enter Second Key: KEY2

Encrypted Cipher Text: 4B454C4F4F
Decrypted Plaintext: HELLO

-----
Process exited after 15.25 seconds with return value 0
Press any key to continue . . .

```
- Compiler Output:**

```

- Errors: 0
- Warnings: 0
- Output Filename: D:\CSA5112 - Cryptography\LAB PROGRAMS\DES.cpp
- Output Size: 323.2841796875 KiB
- Compilation Time: 0.38s

```